Computer Science (CS)70
Network Fundamentals and Architecture
Section 1750
Fall 2025 Semester Sep/02 To Dec/23
This class is ONLINE using Canvas

Transfer: CSU
Units: 3
Advisory: One programming course.

Instructor:	Mehrak Amirfazlian BSc, MSc, AWS Educator, CCNA Instructor, Network security Instructor, A+
Office hours:	7:00 p.m. to 7:48 p.m. Thursdays Via Zoom or by appointment
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Textbook(s)

Networking Essentials: CompTIA Network+ N10-008 Textbook (6th Edition) by Jeffrey S. Beasley and Piyasat Nilkaew.
SMC has an active, institution-paid subscription to O'Reilly Online Learning through its library database and students can log in with their SMC credentials to obtain complimentary access to this e-book.

CompTIA Network+ CertMaster Learn
SMC is covering the cost of student access to this platform. CompTIA Network+ CertMaster Learn is provided at no cost to students. The link to the platform and its assigned content will be available in Canvas.

Materials
This course requires students to have access to a computer, high-speed Internet, and an e-mail account.

Note
This class requires students to work online independently each week. No on-campus meetings will be held.

Netiquette

Communicating online with one another entails certain rules which are not limited to when and where to communicate, but also how to communicate. Please go over the rules for online

learning by visiting albion.com site. Students are expected to follow the ground rules in this class.

Course Description

This course offers a broad introduction to networking concepts and analyzes different network architectures. Introductory topics include network topologies, media and signaling, protocols, addressing, and distributed networks. The varied ways to connect computers are explored as are the resulting architectures. The course explores subnetting, both physical and virtual and internetworks are constructed in the lab. Server programs are introduced to demonstrate their signature socket-API structure. Specific real-world services such as the Apache web server, BIND name server, NFS and Samba file system servers, DHCP address server, and others are discussed.

Course Objectives / Student Learning Outcomes- SLO

1. Explain the essential network components.
2. Select appropriate network topologies for a given environment.
3. Compare and contrast different network media.
4. Understand and explain the function of packets and communication protocols.
5. Explain the limitations and advantages of different network architectures.
6. Install and configure network applications. Assessed by projects, quizzes and final exams.
7. Evaluate and select different network architectures for custom designed environments.

Important dates

Last date to drop this class without a “W”	September 28, 2025
Last date to drop with a “W”	November 23, 2025

For updates on class schedules and important dates visit https://www.smc.edu/academics/classes/

Please remember, **if you wish to drop a class you need to drop it yourself before the deadline** specified above to avoid a “W’ on your academic record. Effective July 1, 2012, students now have just 3 attempts to pass a class. If a student gets a "W" or grade of "D", "F", “I”, or "NP" in a class, that will count as an attempt. A student’s past record of course attempts district-wide will also be considered.

Grading Criteria		Grading scale	
Labs	40%	A	90% to 100%
Reading-Discussion	15%	B	80% to 89%
Quizzes	20%	C	70% to 79%
Final Exam	25%	D	60% to 69%
		F	less than 60%
Total:	100%		

Labs: There will be various labs using packet tracer and/or TestOut Simulation platform.
Reading/Discussion: There are readings and assignments that you need to complete on a weekly basis. Discussion is graded.
Weekly Quiz: For each chapter.
Final Exam: It is comprehensive. It will be available for you to take at the end of the course.

Please check Canvas for any changes on the due dates. Most, if not all, assignments are due the week they are assigned, by Sunday 11:59 p.m. PST. You have approximately seven days to complete all assigned work for that week. If technical issues delay your submission, please inform me immediately. No late submissions are accepted unless there is a justifiable reason. Please check Canvas for all assignments (i.e., labs , exams, discussion).

Incomplete Policy

Students will not be given an incomplete grade in the course without sound reason and documented evidence. In any case, for a student to receive an incomplete, he or she must be passing and have completed a significant portion of the course.

Attendance

Only students who have been admitted to Santa Monica College and are in approved active status may attend classes. If the student is inactive for more than a week (if an 8-week session) or two weeks (if 16-week session), the instructor may exclude the student from the class. Inactive means not attending the class if it is a hybrid or traditional course. For online class, it means that you have not submitted assignments due that week and/or not logged in to the course website often to review course content, reading material, discussions. Attendance in class is expected and necessary for maximum credit. In case of absence, it is your responsibility to find out what material was covered and what assignments were given.
Please note that you are responsible for all material presented and discussed in every class meeting. Please visit https://www.smc.edu/academics/classes/college-catalog/documents/2022-2023/22-23-SMCcatalog-AcadPolicies.pdf for detailed information on SMC policies.

Course Schedule

This course schedule is intended **as a guide** and is subject to change.

Due Dates: All course work is posted on canvas including the due dates. Assigned work must be in before or on the designated due date.

WK (Date)	CompTIA Network+ CertMaster Learn /Readings	Networking Essentials: Beasley and Nilkaew/ Readings
01 (Sep 02)	Ch.1 Explaining Network Topologies	Introduction to Computer Networks Ch.1
02 (Sep 08)	Ch.2 Supporting Cabling and Physical Installations	Physical Layer Cabling Ch.2 & Ch.3
03 (Sep 15)	Ch.3 Configuring Interfaces and Switches	Interconnecting the LANs Ch.5 Introduction to Switch Configuration Ch.8
04 (Sep 22)	Ch.4 Configuring Network Addressing	TCP/IP Ch.6
05 (Sep 29)	Ch.5 Configuring Routing and Advanced Switching	Introduction to Router Configuration Ch.7 Routing Protocols Ch.9
06 (Oct 06)	Ch.6 Implementing Network Services	Managing the Network Infrastructure Ch.10
07 (Oct 13)	Ch.7 Explaining Application Services	Managing the Network Infrastructure Ch.10
08 (Oct 20)	Ch.8 Supporting Network Management	Managing the Network Infrastructure Ch.10
09 (Oct 27)	Ch.9 Explaining Network Security Concepts	Network Security Ch.11
10 (Nov 03)	Ch.10 Applying Network Security Features	Network Security Ch.11
11 (Nov 10)	Ch.11 Supporting Network Security Design	Network Security Ch.11
12 (Nov 17)	Ch.12 Configuring Wireless Networks	Wireless Networking Ch.4
13 (Nov 24)	Ch.13 Comparing Remote Access Methods	Wireless Networking Ch.4
14 (Dec 01)	Ch.14 Summarizing Cloud Concepts	Cloud Computing and Virtualization Ch.12
15 (Dec 08)	Ch.14 Summarizing Cloud Concepts Review, CompTIA Network Practice Materials	Codes and Standards Ch.13
16 (Dec 15)	Final Exam Week: Final Exam Comprehensive	
17 (Dec 22)	End of semester Dec 23	End of semester Dec 23.

Recording devices in classroom

Section 78907 of the California Education Code prohibits the use of any electronic audio or video recording devices, without prior consent of the instructor (including cell phones, laptops, MP3 players, and more) in classroom and/or via Zoom.

Disabled Student Programs and Services (DSPS)

Santa Monica College students with verified disabilities who are requesting academic accommodations should let the instructor know as soon as possible, since the accommodations are not retroactive, this should be done within the first two weeks of the class.
Please contact Center for Students with Disabilities (CSD, also known as DSPS) at 310-434-4265 or use their email dsps@smc.edu to find out the required procedure. The Center for Students with Disabilities is located on the first floor of the Student Services Center.

Rules of Conduct & Academic Integrity (Plagiarism)

Students are expected not to violate the Academic Integrity rules. Violation of Academic Integrity includes but are not limited to the following actions: plagiarism, self-plagiarism, fabrication, deception, cheating, and sabotage, working together on an assignment, paper, or project when the instructor has specifically stated students should not do so, submitting the same term paper to more than one instructor, or allowing another individual to assume one's identity for the purpose of enhancing one's grade. All work you submit must be your own individual work. https://www.smc.edu/academics/classes/2022-23/spring-2023/information-policies.php

Privacy & FERPA

The Family Educational Rights and Privacy Act (FERPA) (20 U.S.C. § 1232g; 34 CFR Part 99) is a federal law that protects the privacy of student education records. The law applies to all schools that receive funds under an applicable program of the U.S. Department of Education. Pierce Community College is subject to FERPA. Generally, schools must have written permission from the student in order to release any information from a student's education record. **Discussing information such as enrollments, prerequisites, and grades over non-stduent.laccd.edu email constitutes a release of information from a student's education record without written permission and thus, a violation of Federal Law.** Information such as enrollments, prerequisites, grades, etc. may be discussed with the student only using *@student.laccd.edu email. Emails sent from non-student.laccd.edu email accounts will be returned unanswered. https://www2.ed.gov/policy/gen/guid/fpco/ferpa/index.html Please do not call or email from a non *.student.laccd.edu email account.

Student Health Center

Medical and mental health services are available to all currently enrolled students at SMC. Please call 310-434-4262 for further information https://www.smc.edu/student-support/health-wellbeing/health-services/

Student Services Assistance and Support

- Please visit https://www.smc.edu/student-support/ for comprehensive list of resources.

- For **Online Tutoring** and more student services visit their site at: https://www.smc.edu/student-support/direct-connect/teaching-learning-growing/tutoring.php

Emergency Preparedness

The safety of students at SMC is a priority. Please visit the following site for more information: https://www.smc.edu/administration/emergency-preparedness/index.php.